

RESEARCH FOCUS	Intersection of AI and statistics across Representation Learning, Interpretability, and LLMs	
CONTACT	taehyo97@gmail.com 917-815-6831	New York, NY
EDUCATION	New York University , New York, NY Ph.D. in Biostatistics (Committee: Hai Shu, Yang Feng, Wen Zhou) M.S. in Computer Science	Expected Spring 2027 2022
	University of Toronto , Toronto, ON B.A.Sc in Computer Engineering, Biomedical Engineering minor	2020
PUBLICATIONS	Kim, T., Shu, H., Jia, Q., de Leon, M. J. DeepFDR: A Deep Learning-based False Discovery Rate Control Method for Neuroimaging Data. (2024). <i>AISTATS</i> , 238: 946–954. Runner-up Winner of JSM Student paper competition [Paper] [Code]	
	Kim, T., Jia, Q., de Leon, M. J., Shu, H. A False Discovery Rate Control Method Using a Fully Connected Hidden Markov Random Field for Neuroimaging Data. (2026). <i>Medical Image Analysis</i> , 104158. [Paper] [Code]	
	Kim, T., Eads, A., McFee, B., McAllister, T., Shu, H. Self-Supervised Learning on Lingual Ultrasound Video Encodes Clinically Meaningful Articulatory Structure of Rhotic Production in Residual Speech Sound Disorder. (2026). <i>MICCAI</i> , 16880: 570–580. [Paper] [Code]	
	Chen, Y., Kim, T., Shu, H., Feng Y. TabSODA: Tabular Diffusion Imputation Framework with Automatic Skip Detection and Ordinal Awareness. (2026). <i>Statistics and Trustworthy AI for Cross Domain Acceleration</i> , In Press. [Paper] [Code]	
	Tang, T., Chen, Y., Kim, T., Shu, H. UKAN-EP: Enhancing U-KAN with Efficient Attention and Pyramid Aggregation for 3D Multi-Modal MRI Brain Tumor Segmentation. (2025). <i>BMC Medical Imaging</i> , 25, 517. [Paper] [Code]	
	Hosseini, M. S., Bejnordi, B. E., Trinh, V. Q., Chan, L., Hasan, D., Li, X., Yang, S., Kim, T. et al. Computational Pathology: A Survey Review and the Way Forward. (2024). <i>Journal of Pathology Informatics</i> , 15, 100357. [Paper]	
SUBMITTED OR UNDER REVISION	Eads, A., Kim, T., et al. Identifying subtypes of rhotic distortions using self-supervised learning on lingual ultrasound data. (2026). <i>Minor Revision at Journal of Speech, Language, and Hearing Research</i> .	
	Kim, T., Shu, H. SKARF: LLM-based Knowledge Augmentation of Random Forests for Enhanced Survival Prediction. (2026). <i>Submitted to TLMR</i> .	
	Li, B., Kim, T., Shu, H. DePHI: Dynamically Evolving System Framework for PHI-Safe and Auditable LLM Generation. (2026). <i>Submitted to Neurips 2026</i> .	
	Lai, A., Kim, T., Dahlen, A., Lomas, T. A Global Understanding of Work Enjoyment and Human Wellbeing. (2025).	

Round 1 Review at Social Indicators Research.

Kim, H., Cardoso, D. d. M., Kayahara, G. M., Kim, T. et al. Phenotyping Neuropathic-Like Pain and Functional Impairment in Head and Neck Cancer Across Disease and Treatment States using a Novel Orofacial Sensory Phenotyping Questionnaire. (2026).
Submitted to British Journal of Anesthesia.

CURRENT PROJECTS	Kim, T., Shu, H. L0-IPLS: An L0 Penalized Sparse Canonical Correlation Analysis Method with Application to Imaging-Omics Data. <i>In preparation for Biometrics.</i>	
MEMBERSHIP AND AFFILIATIONS	Member, American Statistical Association	2022 - present
	Member, Eastern North American Region of the International Biometric Society	2022 - present
	Member, Korean International Statistical Society	2024 - present
	Reviewer, Medical Image Analysis	September 2026
	Reviewer, Computational Statistics & Data Analysis	August 2026
	Reviewer, IEEE International Conference on Data Mining (ICDM)	July 2026
	Reviewer, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)	Mar 2026
	Reviewer, The American Statistician	Dec 2025
	Reviewer, The American Statistician	Aug 2025
AWARDS	FDA-OCE-ASA Oncology Educational Fellowship, FDA	2026
	JSM Student Paper Award, ASA Statistics in Imaging	2024
	DataFest Finalist, ENAR	2024
	Student Travel Award, American Statistical Association	2024
	PhD Fellowship Award, New York University	2022
	Certificate for Artificial Intelligence Engineering, University of Toronto	2020
	International Summer Research Award, University of Toronto	2019
	Gold Award, The Duke of Edinburgh's Award	2015
CONFERENCE PRESENTATION	"A False Discovery Rate Control Method Using a Fully Connected Hidden Markov Random Field for Neuroimaging Data"	
	Invited Oral, ASA SLDS, Brooklyn, NY	November 2026
	"Self-Supervised Learning on Lingual Ultrasound Video Encodes Clinically Meaningful Articulatory Structure of Rhotic Production in Residual Speech Sound Disorder"	
	Poster, MICCAI, Strasbourg, France	September 2026
	"A Global Understanding of Work Enjoyment and Human Wellbeing"	
	Poster, Joint Statistical Meetings, Nashville, TN	August 2025
	"Causal Determinants of Blood Pressure Control among US Adults with Hypertension: A Data Driven Causal Graphical Learning, NHANES 2013 to 2023"	
	Poster, ENAR Spring Meeting, New Orleans, LA	March 2025
	"DeepFDR: A Deep Learning-based False Discovery Rate Control Method for Neuroimaging Data"	
	Invited Oral, Joint Statistical Meetings, Portland, OR	August 2024
	"DeepFDR: A Deep Learning-based False Discovery Rate Control Method for Neuroimaging Data"	
	Poster, International Conference in Artificial Intelligence and Statistics, Valencia, Spain	May 2024

	“Enhancing AI-based Speech Therapy through Acoustic to Articulatory Mapping” Poster, AI Research Symposium: Bridging AI Innovation and Societal Impact New York, NY	April 2024
	“Machine Learning-driven Risk Factor Identification on Post-2013 Blood Pressure Control Decline in Hypertensive Populations” Oral, ENAR Spring Meeting, Baltimore, MD	March 2024
TEACHING EXPERIENCE	Teaching Assistant, New York University, New York, NY Statistical Methods in Genomics and Bioinformatics (GPH-GU 2378) Survey Design, Analysis, and Reporting (GPH-GU 2387) Applied Bayesian Analysis in Public Health (GPH-GU 2272/3372) Applied Survival Analysis (GPH-GU 2368/3368) Statistical Inference (GPH-GU 3225)	Spring 2026 Fall 2025 Fall 2024 Spring 2024 Fall 2023
	Graduate Student Mentor, New York University, New York, NY Pathways into Quantitative Aging Research Summer Program Pathways into Quantitative Aging Research Summer Program	Summer 2024 Summer 2022
PROFESSIONAL EXPERIENCE	Machine Learning Research Intern Dandy - Proposed a preference learning framework for 3D dental crown quality using VAE and Bradley Terry ranking head, improving the unguided manufacturing rate from 36% to 66%. - Represented 3D data in a topology preserving 2D latent space, accelerating training by 4x. Statistical Machine Learning Researcher NYU School of Global Public Health <u>Thesis title</u>: Advancing statistical machine learning for Alzheimer’s disease research - Designed computationally efficient, spatial false discovery rate control methods for high dimensional neuroimaging, adapting unsupervised W-Net and hidden Markov random fields. - Formulated an L0 penalized sparse canonical correlation analysis method for interpretable multi-modal representation learning with application to imaging-omics data. - Proposed knowledge augmented, interpretable survival modeling framework utilizing LLMs. NYU Steinhardt School of Culture, Education, and Human Development - Developed a self-supervised learning pipeline for lingual ultrasound video analysis. NYU College of Dentistry - Conducted data curation and statistical analysis for clinical dental research. NYU Courant Institute School of Mathematics, Computing, and Data Science - Built an Apple Watch system to detect, track, and help alleviate pediatric anxiety attacks. FDA-OCE-ASA Oncology Educational Fellow U.S. Food and Drug Administration (FDA) - Evaluated oncology clinical trial evidence and regulatory decision-making through FDA-led case studies and a mock Oncologic Drugs Advisory Committee. Deep Learning Researcher Biofeedback Intervention Technology for Speech Lab, NYU Steinhardt - Presented the first self-supervised learning framework that focuses on within-rhotic articulatory structure in American English from lingual ultrasound video of children with residual SSD. - Integrated speaker-adversarial training that outperforms BYOL, VideoMAE, and RESA by 8% accuracy and 0.194 MCC on within-phoneme structure classification.	Summer 2026 New York, NY Sept 2020 - present New York, NY Oct 2025 - May 2026 Remote Sept 2024 - May 2026 New York, NY

Statistical Fellow Summer 2024
Biostatistical Collaboration and Consultation Core, NYU GPH New York, NY
• Consulted clients to define statistical analysis plans and supported manuscript development.

Research Engineer Intern Summer 2019
N.I Institute for Health, National University of Singapore Kent Ridge, Singapore
• Assembled a parallel-distributed high-performance cluster system via HTCondor, Unix, and MATLAB to enable high throughput analysis of 76TB neural coding data.

Software Developer Intern May 2018 - May 2019
Seiko Epson Markham, ON
• Developed object tracking ground truth generation system by calibrating OptiTrack Motion Capture and Unity3D in real time.
• Implemented the Windows Software Development Kit (SDK) supporting Epson's augmented reality smart-glasses.

SKILLS

Programming: Python, PyTorch/JAX/Tensorflow, C/C++/C#, R, SQL, Java
MLOps: MLflow, Flyte, Vertex AI, Spark, Docker, Kubernetes, Git, BigQuery, LangChain